Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION Model Question Paper

Fourth SEMESTER

Discipline Specific Core (DSC) – MG4DSCPHY201 – ELECTROMAGNETIC THEORY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **3** marks

1. Define electric field intensity. Write its SI unit [U] / [CO1]
2. State Gauss's law in electrostatics. [U] / [CO1]
3. Define electric potential. How is it related to electric field? [U] / [CO2]
4. Write Poisson's equation and mention one application [E] / [CO2]
5. State Lorentz force law. [U] / [CO3]
6. What is magnetic vector potential? Write the relation between B and A [U] / [CO3]
7. Define polarization in dielectrics. [U] / [CO5]
8. What is electric displacement vector D? [U] / [CO5]
9. State Faraday's law of electromagnetic induction. [U] / [CO6]
10. Write any two Maxwell's equations in differential form. [K] / [CO7]
11. State Ampere's circuital law [U] / [CO4]

12. Mention any two applications of Ampere's law [A] / [CO4]

[10x3 = 30]

Part B

Short Essay Type Questions

Answer any **8** questions

Each question carries **5** marks

13. Obtain the expression for electric field due to a continuous charge distribution' [K] / [CO2]

14. Explain divergence and curl of electrostatic field with physical significance [U] / [CO1]

15. Obtain Laplace's equation and discuss its importance in electrostatics [E] / [CO2]

16 Compare electrostatics and magnetostatics with suitable examples. [U]/[[CO3][C04]

17. Write Maxwell's equations in free space and explain their physical significance. [U] / [CO7]

18. Using Biot–Savart law, derive the magnetic field due to a long straight current-carrying conductor. [U] / [CO3]

19 A long solenoid has 1000 turns per meter and carries a current of 2A. Find the magnetic field inside the solenoid. [U] / [CO4]

20. Explain electric displacement and Gauss's law in the presence of dielectrics. [U] / [CO5]

21. A circular coil with 50 turns and area 0.02 m^2 is placed in a magnetic field that changes uniformly from 0.5 T to zero in 0.1 s. Calculate the induced emf in the coil. [U] / [CO6]

22 Derive the continuity equation and explain its physical significance. [U]/[CO6]

[8x5 = 40]